PROJECT SCHEDULE, CRITICAL PATH ANALYSIS AND BUDGET ESTIMATION

ShopEase Online Store: Development of an E-commerce Web Application

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Schedule and budget model : ShopEase_Schedule_Budget_Model.xlsx

ABSTRACT

This document is the schedule and cost plan for the ShopEase online store project, whose charter, WBS and WBS dictionary were prepared in Assignment 1. The 25 work packages of the WBS were converted into 26 schedule activities. Durations were estimated with three-point (PERT) estimates, and the activities were sequenced with finish-to-start, start-to-start and finish-to-finish dependencies, including leads and lags.

An activity-on-node network diagram was drawn and the Critical Path Method (forward and backward pass) was applied. The project takes 108 working days. The critical path is A - C - D - I - K - L - P - S - T - V - W - X - Y - Z, giving go-live on 27 Jan 2027 and closure on 16 Feb 2027, which matches the charter. Total float and free float were calculated for every activity, resource loading was checked and levelled within float, and crashing and fast-tracking options were compared.

The budget was estimated bottom-up from effort and daily rates for each activity and rolled up to the WBS control accounts. The work package total is Rs 10,17,800; a contingency reserve of Rs 1,02,000, based on the expected monetary value of the project risks, gives a cost baseline of Rs 11,19,800, and a management reserve of Rs 51,000 gives a total budget of Rs 11,70,800. The cost baseline was time-phased by week and plotted as an S-curve for both early and late start schedules. All calculations are also built as formulas in an Excel workbook, so the critical path, dates, floats, budget and S-curve recalculate when a duration, dependency, rate or date is changed.

Keywords: Project Schedule, Network Diagram, Precedence Diagramming Method, Critical Path Method, Float, PERT, Crashing, Fast Tracking, Cost Baseline, Contingency Reserve, Management Reserve, S-Curve.

1. INTRODUCTION

1.1 Background

ShopEase is a clothing and lifestyle store in Pune that is building its own online store to replace ordering through WhatsApp messages. Assignment 1 authorised the project through a project charter and defined the scope with a WBS of 8 control accounts and 25 work packages. This assignment takes the scope baseline and produces the next two baselines of the project management plan: the **schedule baseline** and the **cost baseline**.

1.2 Objectives

1. To define the schedule activities from the WBS work packages and estimate their durations.
2. To sequence the activities with the correct dependency types, leads and lags.
3. To draw the network diagram and find the critical path and float with the Critical Path Method.
4. To build the project schedule (Gantt chart and milestones) and check resource loading.
5. To analyse schedule compression options (crashing and fast tracking).
6. To estimate the budget bottom-up, aggregate it into a cost baseline with reserves and plot the S-curve.
7. To build a dynamic model that recalculates the schedule and budget when inputs change.

1.3 Planning Basis and Assumptions

- **Calendar:** 5 working days per week (Monday to Friday). Project day 0 is Tuesday 01 Sep 2026; week 1 is the week starting Monday 31 Aug 2026. Holidays: Ganesh Chaturthi (14 Sep), Gandhi Jayanti (2 Oct), Dussehra (20 Oct), Diwali (9 to 11 Nov), Christmas (25 Dec), office closure from 28 Dec 2026 to 1 Jan 2027 and Republic Day (26 Jan 2027), 13 working days in total.
- **Units:** all durations, leads, lags and floats are in working days. ES and EF are day numbers counted from day 0; an activity with ES = 5 and duration 5 occupies days 5 to 9 and has EF = 10.
- **Team:** Project Manager, Business Analyst, UI/UX Designer, 2 Full-stack Developers, QA Engineer and DevOps Engineer, as in the charter. Developer 1

builds the customer storefront chain and Developer 2 builds the admin chain; these resource sequences are included as dependencies.

- **Monitoring and status reporting (WBS 1.1.3)** is a level-of-effort activity that runs from the start of planning until the end of hypercare. It has no logic of its own and is not drawn in the network, but its cost is included in the budget and S-curve.
- Rates and non-labour costs are the same as in the WBS dictionary, so the budget in this document traces directly to Assignment 1.

2. ACTIVITY DEFINITION AND DURATION ESTIMATES

2.1 From Work Packages to Activities

Each work package of the WBS became one activity, except work package 1.8.3 (Go-Live and Hypercare), which was split into X (go-live preparation and deployment) and Y (hypercare) because they have different logic and the go-live date is the start of Y. Activity IDs A to Z are used in the network diagram; the WBS code links every activity back to the WBS dictionary.

2.2 Three-Point Estimates

The team gave an optimistic (O), most likely (M) and pessimistic (P) duration for each activity. The expected duration uses the PERT (beta) formula and is rounded to whole working days:

$$t_e = (O + 4M + P) / 6 \qquad \text{standard deviation } \sigma = (P - O) / 6$$

Example, activity I (Product catalogue and search): O = 12, M = 14, P = 18, so t_e = (12 + 56 + 18) / 6 = 14.33, rounded to 14 days, and σ = (18 - 12) / 6 = 1.00 day. Hypercare (Y) is fixed at 10 days by the charter, so O = M = P.

ID	Activity	WBS	O	M	P	t_e	σ	Dur
A	Project initiation and charter	1.1.1	4	5	6	5.00	0.33	**5**
B	Project planning (baselines)	1.1.2	4	5	9	5.50	0.83	**6**
C	Requirements specification	1.2.1	8	10	12	10.00	0.67	**10**
D	UI/UX design and prototype test	1.2.2	12	15	18	15.00	1.00	**15**
E	Architecture and database design	1.2.3	8	10	12	10.00	0.67	**10**
F	Cloud hosting and environments	1.6.1	5	7	9	7.00	0.67	**7**
G	CI/CD pipeline	1.6.2	4	5	6	5.00	0.33	**5**
H	User accounts and OTP login	1.3.2	6	8	10	8.00	0.67	**8**
I	Product catalogue and search	1.3.1	12	14	18	14.33	1.00	**14**
J	Product and inventory admin	1.5.1	8	10	12	10.00	0.67	**10**
K	Shopping cart and checkout	1.3.3	12	14	17	14.17	0.83	**14**

ID	Activity	WBS	O	M	P	t_e	σ	Dur
L	Payment gateway integration	1.4.1	8	10	14	10.33	1.00	**10**
M	Order management and returns	1.5.2	8	10	14	10.33	1.00	**10**
N	Shipping and courier integration	1.4.2	5	6	8	6.17	0.50	**6**
O	GST invoice generation	1.4.3	4	5	7	5.17	0.50	**5**
P	Order tracking and notifications	1.3.4	6	8	10	8.00	0.67	**8**
Q	Sales reports dashboard	1.5.3	5	6	7	6.00	0.33	**6**
R	Security hardening and backups	1.6.3	8	10	12	10.00	0.67	**10**
S	Functional and integration testing	1.7.1	16	20	24	20.00	1.33	**20**
T	Performance and security testing (VAPT)	1.7.2	6	8	12	8.33	1.00	**8**
U	Product data migration	1.8.1	10	12	16	12.33	1.00	**12**
V	User acceptance testing	1.7.3	4	6	8	6.00	0.67	**6**
W	Staff training and user manual	1.8.2	6	8	10	8.00	0.67	**8**
X	Go-live preparation and deployment	1.8.3	3	4	5	4.00	0.33	**4**
Y	Hypercare (2 weeks)	1.8.3	10	10	10	10.00	0.00	**10**
Z	Project closure	1.1.4	4	5	6	5.00	0.33	**5**

Table 1: Activity list with three-point estimates (durations in working days)

3. ACTIVITY SEQUENCING

3.1 Dependency Types Used

The schedule uses the Precedence Diagramming Method (PDM), where activities are nodes and dependencies are arrows. Three of the four PDM relationship types are used:

Type	Meaning	Example in this project
FS (finish-to-start)	Successor cannot start until predecessor finishes	K Cart and checkout -> L Payment gateway
SS (start-to-start)	Successor cannot start until predecessor has started (plus lag)	F Hosting SS+3 -> G CI/CD: pipeline starts 3 days after hosting setup starts
FF (finish-to-finish)	Successor cannot finish until predecessor has finished (plus lag)	P Tracking FF+4 -> S Testing: testing ends 4 days after the last feature is finished
SF (start-to-finish)	Successor cannot finish until predecessor starts	Not needed in this project

Table 2: PDM relationship types

A **lag** is a waiting time added to a relationship (for example J FS+15 -> U, because suppliers need three weeks to send product data after the bulk upload is ready). A **lead** is a negative lag that lets the successor start early (C FS-2 -> E: architecture starts 2 days before the SRS is formally finished, using the final draft).

Dependencies were also classified as **mandatory** (hard logic, for example payment needs checkout), **discretionary** (preferred logic that can be changed for fast tracking), **external** (depends on suppliers or vendors) and **resource** (the same developer cannot do two activities at once).

3.2 Dependency Register

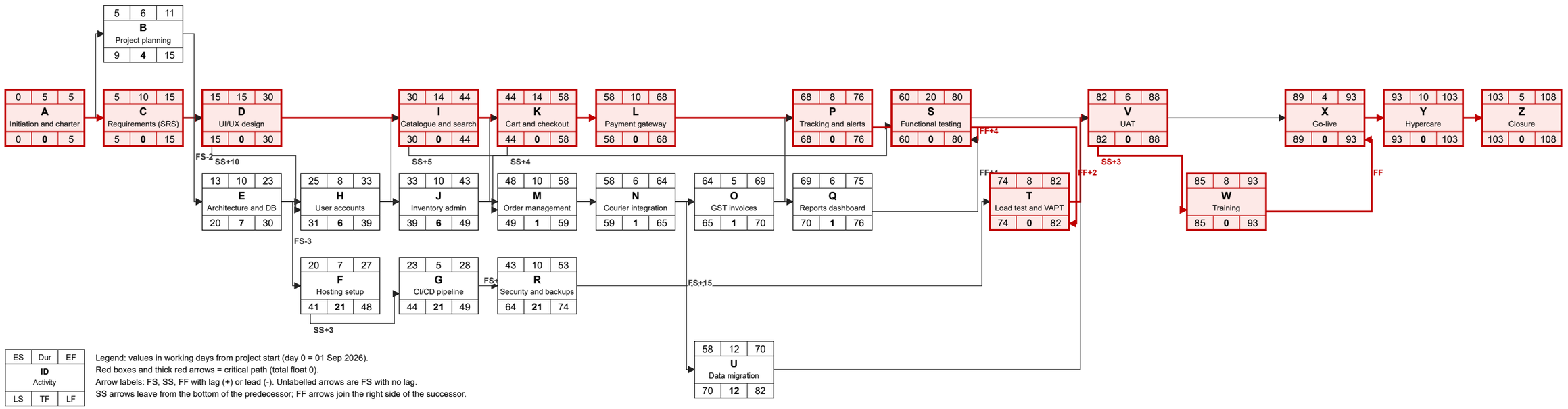
No	Pred.	Succ.	Type	Lag	Category	Reason
1	A	B	FS	0	Mandatory	Baselines need the approved charter
2	A	C	FS	0	Mandatory	Requirements work is authorised by the charter
3	C	D	FS	0	Mandatory	Screens are designed from approved user stories
4	B	D	FS	0	Discretionary	Design starts only after the plan is baselined
5	C	E	FS	-2	Discretionary	Lead of 2 days: architecture starts on the final SRS draft
6	E	F	FS	-3	Discretionary	Lead of 3 days: hosting set up once the stack is chosen
7	F	G	SS	+3	Mandatory	SS + 3: pipeline needs the staging server, not all environments
8	E	H	FS	0	Mandatory	Needs database schema and API design
9	D	H	SS	+10	Discretionary	SS + 10: login screens are designed first, so coding starts early
10	D	I	FS	0	Mandatory	Catalogue pages need the signed-off design
11	E	I	FS	0	Mandatory	Needs database schema
12	H	J	FS	0	Resource	Developer 2 moves from accounts to admin
13	I	K	FS	0	Mandatory	Cart needs products; also Developer 1 sequence
14	H	K	FS	0	Mandatory	Checkout needs logged-in customers and addresses
15	K	L	FS	0	Mandatory	Payment is the last step of checkout; Developer 1 sequence
16	K	M	SS	+4	Discretionary	SS + 4: order screens start once the order model in checkout is fixed
17	J	M	FS	0	Resource	Developer 2 sequence
18	M	N	FS	0	Mandatory	Shipments are created from processed orders
19	N	O	FS	0	Resource	Developer 2 sequence
20	L	P	FS	0	Mandatory	Notifications are triggered by payment status
21	N	P	FS	0	Mandatory	Tracking status comes from the courier API
22	O	Q	FS	0	Resource	Developer 2 sequence; reports use invoice data
23	G	R	FS	+15	Discretionary	FS + 15: hardening done once the main features are deployed
24	I	S	SS	+5	Discretionary	SS + 5: testing starts with the first finished feature
25	P	S	FF	+4	Mandatory	FF + 4: testing ends 4 days after the last feature
26	Q	S	FF	+4	Mandatory	FF + 4: same as above for the admin side
27	R	T	FS	0	Mandatory	Audit is done on the hardened system
28	S	T	FF	+2	Mandatory	FF + 2: load and security tests finish after functional testing
29	J	U	FS	+15	External	FS + 15: bulk upload ready, then 3 weeks for supplier data and photos
30	S	V	FS	0	Mandatory	UAT on a tested build

No	Pred.	Succ.	Type	Lag	Category	Reason
31	T	V	FS	0	Mandatory	No open critical VAPT findings before UAT
32	U	V	FS	0	Mandatory	UAT uses real product data
33	V	W	SS	+3	Discretionary	SS + 3: training uses the UAT environment
34	V	X	FS	0	Mandatory	Go-live needs UAT sign-off
35	W	X	FF	0	Mandatory	FF: staff must be trained before the site goes live
36	X	Y	FS	0	Mandatory	Hypercare starts at go-live
37	Y	Z	FS	0	Mandatory	Closure after hypercare

Table 3: Dependency register (37 relationships)

There are no circular dependencies: the activities are listed in an order where every predecessor appears before its successor (a topological order), which is also what allows the forward pass to be calculated row by row in the Excel model.

4. PROJECT NETWORK DIAGRAM



Legend: values in working days from project start (day 0 = 01 Sep 2026).
Red boxes and thick red arrows = critical path (total float 0).
Arrow labels: FS, SS, FF with lag (+) or lead (-). Unlabelled arrows are FS with no lag.
SS arrows leave from the bottom of the predecessor; FF arrows join the right side of the successor.

Figure 1: Activity-on-node network diagram with ES, duration, EF, LS, total float and LF for every activity

5. CRITICAL PATH METHOD

5.1 Forward Pass

The forward pass calculates the early start (ES) and early finish (EF) of each activity, starting with ES = 0 for activity A. For each activity, ES is the largest value given by its predecessors:

- FS: ES = EF(pred) + lag
- SS: ES = ES(pred) + lag
- FF: ES = EF(pred) + lag - duration (the successor must not finish before the predecessor finishes + lag)
- EF = ES + duration

Worked examples:

- E (FS-2 from C): ES = EF(C) - 2 = 15 - 2 = **13**; EF = 13 + 10 = 23.
- H (FS from E, SS+10 from D): ES = max(EF(E) = 23, ES(D) + 10 = 15 + 10 = 25) = **25**; EF = 33.
- S (SS+5 from I, FF+4 from P, FF+4 from Q): ES = max(30 + 5 = 35, EF(P) + 4 - 20 = 76 + 4 - 20 = 60, EF(Q) + 4 - 20 = 75 + 4 - 20 = 59) = **60**; EF = 80. The FF link from P controls S, so testing is pushed to finish 4 days after the last storefront feature.
- V (FS from S, T, U): ES = max(80, 82, 70) = **82**.

The largest EF is 108 (activity Z), so the project duration is **108 working days**.

5.2 Backward Pass

The backward pass starts with LF = 108 for the last activity and calculates the late finish (LF) and late start (LS). For each activity, LF is the smallest value given by its successors:

- FS: LF = LS(succ) - lag
- SS: LF = LS(succ) - lag + duration
- FF: LF = LF(succ) - lag
- LS = LF - duration; Total float TF = LS - ES; Free float FF = time the activity can slip without delaying the early start of any successor.

Worked examples:

- Q (FF+4 to S): LF = LF(S) - 4 = 80 - 4 = **76**; LS = 70; TF = 70 - 69 = **1**.
- F (SS+3 to G): LF = LS(G) - 3 + 7 = 44 - 3 + 7 = **48**; LS = 41; TF = 41 - 20 = **21**.

- E (FS-3 to F, FS to H, FS to I): LF = min(LS(F) + 3 = 44, LS(H) = 31, LS(I) = 30) = **30**; LS = 20; TF = **7**.
- U (FS to V): LF = LS(V) = 82, LS = 70, TF = 70 - 58 = **12**. Its free float is also 12 because V's early start (82) is set by T.

5.3 CPM Results

ID	Activity	Dur	ES	EF	LS	LF	TF	FF	Crit.	Start	Finish
A	Project initiation and charter	5	0	5	0	5	**0**	0	Yes	01 Sep	07 Sep
B	Project planning (baselines)	6	5	11	9	15	**4**	4		08 Sep	16 Sep
C	Requirements specification	10	5	15	5	15	**0**	0	Yes	08 Sep	22 Sep
D	UI/UX design and prototype test	15	15	30	15	30	**0**	0	Yes	23 Sep	14 Oct
E	Architecture and database design	10	13	23	20	30	**7**	0		21 Sep	05 Oct
F	Cloud hosting and environments	7	20	27	41	48	**21**	0		30 Sep	09 Oct
G	CI/CD pipeline	5	23	28	44	49	**21**	0		06 Oct	12 Oct
H	User accounts and OTP login	8	25	33	31	39	**6**	0		08 Oct	19 Oct
I	Product catalogue and search	14	30	44	30	44	**0**	0	Yes	15 Oct	04 Nov
J	Product and inventory admin	10	33	43	39	49	**6**	0		21 Oct	03 Nov
K	Shopping cart and checkout	14	44	58	44	58	**0**	0	Yes	05 Nov	27 Nov
L	Payment gateway integration	10	58	68	58	68	**0**	0	Yes	30 Nov	11 Dec
M	Order management and returns	10	48	58	49	59	**1**	0		16 Nov	27 Nov
N	Shipping and courier integration	6	58	64	59	65	**1**	0		30 Nov	07 Dec
O	GST invoice generation	5	64	69	65	70	**1**	0		08 Dec	14 Dec
P	Order tracking and notifications	8	68	76	68	76	**0**	0	Yes	14 Dec	23 Dec
Q	Sales reports dashboard	6	69	75	70	76	**1**	1		15 Dec	22 Dec
R	Security hardening and backups	10	43	53	64	74	**21**	21		04 Nov	20 Nov
S	Functional and integration testing	20	60	80	60	80	**0**	0	Yes	02 Dec	06 Jan
T	Performance and security testing (VAPT)	8	74	82	74	82	**0**	0	Yes	22 Dec	08 Jan
U	Product data migration	12	58	70	70	82	**12**	12		30 Nov	15 Dec
V	User acceptance testing	6	82	88	82	88	**0**	0	Yes	11 Jan	18 Jan
W	Staff training and user manual	8	85	93	85	93	**0**	0	Yes	14 Jan	25 Jan
X	Go-live preparation and deployment	4	89	93	89	93	**0**	0	Yes	20 Jan	25 Jan
Y	Hypercare (2 weeks)	10	93	103	93	103	**0**	0	Yes	27 Jan	09 Feb
Z	Project closure	5	103	108	103	108	**0**	0	Yes	10 Feb	16 Feb

Table 4: Forward pass, backward pass and float (working days); dates are early start and early finish

5.4 Critical Path and Near-Critical Paths

The **critical path** is **A - C - D - I - K - L - P - S - T - V - W - X - Y - Z** with zero total float. It runs through requirements and design, the storefront development chain of Developer 1 (catalogue, checkout, payment, tracking), testing, UAT, training, go-live, hypercare and closure. Any delay on these activities delays go-live day for day.

Path (through)	Total float	Comment
A-C-D-I-K-L-P-S-T-V-W-X-Y-Z	0	Critical path (storefront chain, Developer 1)
...-K-M-N-O-Q-S-...	1	Admin and integration chain, Developer 2 (near-critical)
...-D-H-K-...	6	Accounts before checkout
...-H-J-M-...	6	Inventory admin chain
...-E-I-...	7	Architecture to catalogue
...-J-U-V-...	12	Product data migration
...-E-F-G-R-T-...	21	Infrastructure and security
A-B-D-...	4	Planning before design

Table 5: Main paths and their float

The admin chain M - N - O - Q has only **1 day** of float, so it is near-critical and is tracked as closely as the critical path. Activities B, E, H, J have 4 to 7 days of float, U has 12 days and the infrastructure chain F - G - R has 21 days, which is where the project manager can borrow time or move resources without affecting go-live.

5.5 Schedule Risk (PERT)

The standard deviation of the critical path is the square root of the sum of the variances of the critical activities: $\sigma_{CP} = \sqrt{(\Sigma \sigma^2)}$ = **2.93 days**. The planned dates are the expected (50%) dates. The probability of finishing within 5 working days of the plan is $P(Z \leq 5 / 2.93)$ = **96%**. The near-critical admin chain is not included in this simple PERT calculation, so the real risk is slightly higher, which is one reason for keeping the contingency reserve.

6. PROJECT SCHEDULE

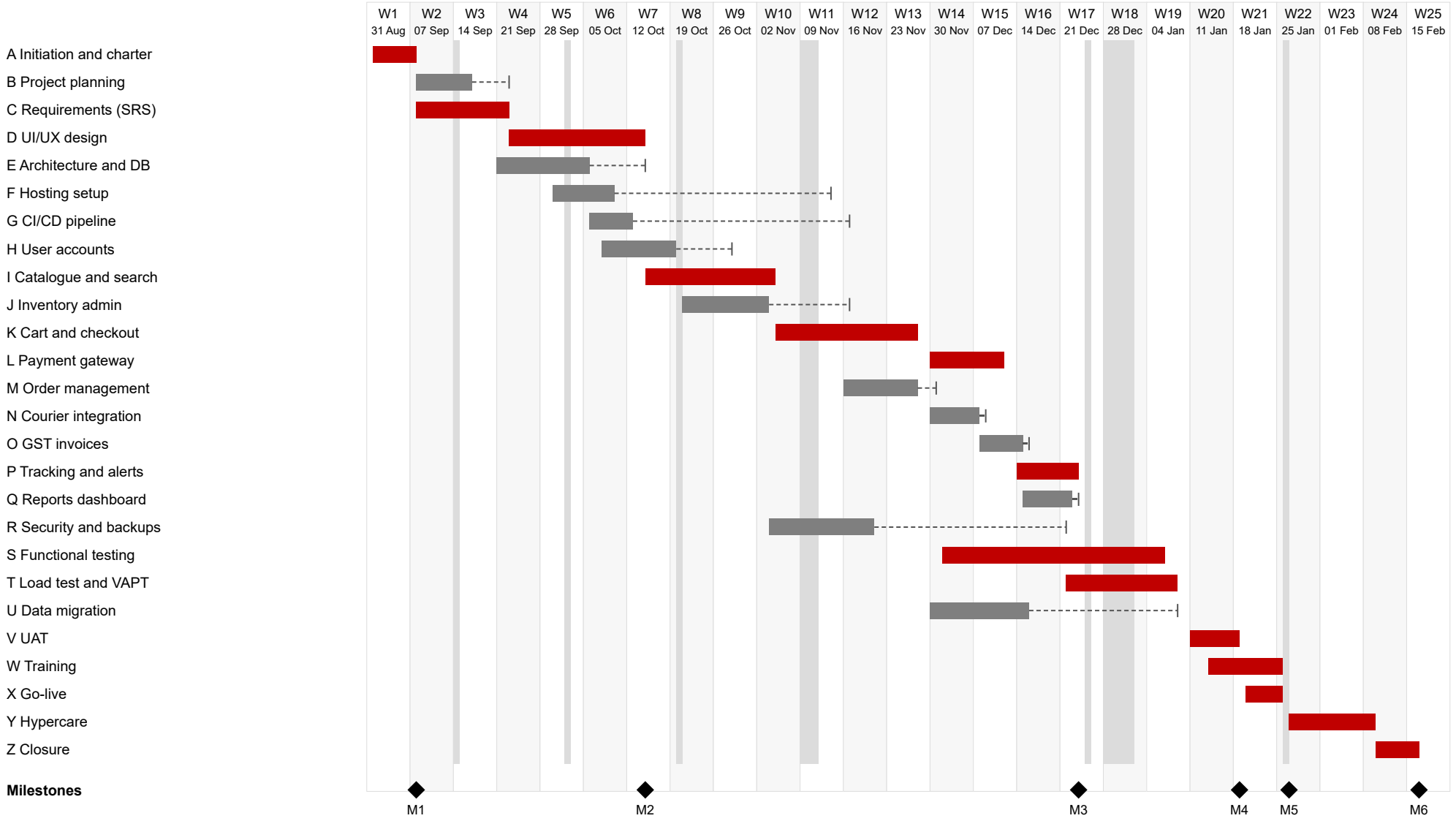


Figure 2: Gantt chart (early start schedule). Red = critical, grey = non-critical, dashed line = total float

6.1 Milestone Schedule

ID	Milestone	Scheduled date	Week	Approved by
M1	Charter approved	07 Sep 2026	W2	Sponsor
M2	Design signed off	14 Oct 2026	W7	Sponsor, Operations Head
M3	Feature complete	23 Dec 2026	W17	Project Manager
M4	UAT signed off	18 Jan 2027	W21	Sponsor
M5	Go-live	27 Jan 2027	W22	Sponsor (go/no-go)
M6	Project closed	16 Feb 2027	W25	Sponsor

Table 6: Milestone schedule

These milestone dates are the ones recorded in the charter (Assignment 1). Go-live is on Wednesday 27 Jan 2027, the day after Republic Day, so the go/no-go meeting is held on 25 Jan 2027. Milestone M3 (feature complete) is reached when the last development activity (P, tracking and notifications) finishes. A change of up to 5 working days in a milestone that does not affect go-live is within the project manager's authority; anything larger goes to the sponsor.

7. RESOURCE ALLOCATION AND LEVELLING

CPM assumes unlimited resources, so the schedule was checked against the team capacity. The developer demand per week was calculated by spreading each activity's developer effort evenly over its duration.

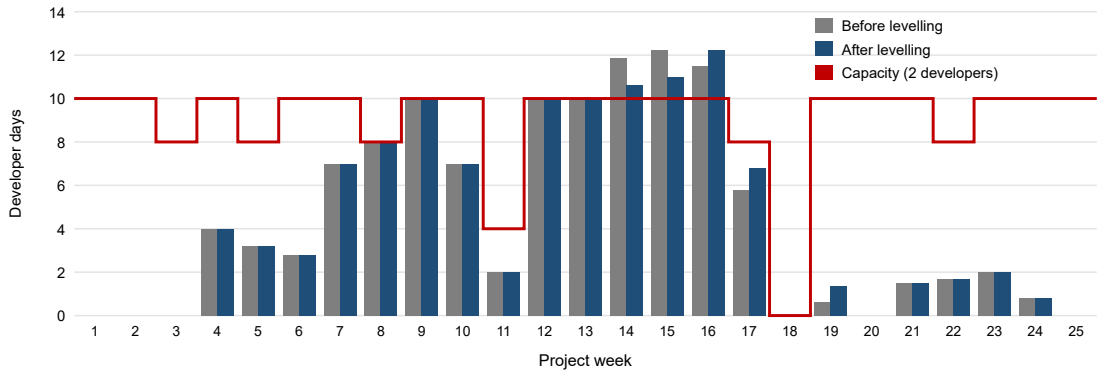


Figure 3: Developer loading per week before and after levelling (capacity = 2 developers x working days)

Before levelling, the developers are over-allocated in weeks 14, 15 and 16 (up to 12.2 developer-days against a capacity of 10), because developer support for data migration (U) and defect fixing during testing (S) overlap with the last feature work. Activity U has 12 days of total float, so it was delayed by 10 days within its float, which moves its developer work out of the busiest weeks. The remaining excess of about 4 developer-days in weeks 14, 15 and 16 is defect fixing inside testing; it is

covered by planned Saturday work (about 2 days per developer, about Rs 13,475 from the contingency reserve) instead of delaying the critical path. Week 11 has only 2 working days because of Diwali, so its capacity is lower. The DevOps engineer is over-allocated in week 6 (hosting F and CI/CD G overlap); G has 21 days of float and is delayed by 4 days. Neither change affects the critical path or the go-live date.

Role	Effort (person-days)	Main weeks of work	Peak load
Project Manager	52	All weeks (about 1.5 days a week for monitoring); other work in W1-W4, W20-W25	W2
Business Analyst	32	W1-W7, W14-W16, W20-W22, W24-W25	W2
UI/UX Designer	15	W4-W7	W6
Developers (2)	112	W4-W17, W19, W21-W24	W16
QA Engineer	30	W14-W17, W19-W24	W19
DevOps Engineer	26	W4-W7, W10-W12, W17, W19, W21-W24	W7 (after levelling)

Table 7: Resource allocation summary

8. SCHEDULE COMPRESSION

If the sponsor asks for an earlier go-live (for example to have the store live well before the Valentine's Day sale), the schedule can be compressed without changing the scope. Only activities on the critical path shorten the project.

8.1 Crashing

Crashing adds resources to critical activities. The **cost slope** = extra cost / days saved shows which activity is cheapest to crash.

ID	Activity	Normal	Crash	Extra cost	Slope (Rs/day)	How
C	Requirements specification	10	8	Rs 9,000	Rs 4,500	Business analyst works 2 Saturdays; PM joins interviews
D	UI/UX design and prototype test	15	12	Rs 18,000	Rs 6,000	Freelance designer for 6 days at Rs 3,000
I	Product catalogue and search	14	11	Rs 31,500	Rs 10,500	Contract developer for 7 days at Rs 4,500
K	Shopping cart and checkout	14	11	Rs 31,500	Rs 10,500	Contract developer for 7 days at Rs 4,500
L	Payment gateway integration	10	8	Rs 18,000	Rs 9,000	Paid integration support package from the payment gateway
P	Order tracking and notifications	8	6	Rs 14,000	Rs 7,000	Developer overtime (2 developers, 2 days)

ID	Activity	Normal	Crash	Extra cost	Slope (Rs/day)	How
S	Functional and integration testing	20	16	Rs 33,600	Rs 8,400	Second QA engineer for 12 days at Rs 2,800
V	User acceptance testing	6	5	Rs 6,000	Rs 6,000	Extra UAT session on a Saturday

Table 8: Crashing options

The activities were crashed one day at a time, always choosing the cheapest critical activity that actually shortens the project, and the CPM was recalculated after each step:

Step	Crashed	Cost	Cumulative cost	New go-live	Critical activities after step
1	C	Rs 4,500	Rs 4,500	25 Jan 2027	A C D I K L P S T V W X Y Z
2	C	Rs 4,500	Rs 9,000	22 Jan 2027	A C D I K L P S T V W X Y Z
3	D	Rs 6,000	Rs 15,000	21 Jan 2027	A C D I K L P S T V W X Y Z
4	D	Rs 6,000	Rs 21,000	20 Jan 2027	A C D I K L P S T V W X Y Z
5	D	Rs 6,000	Rs 27,000	19 Jan 2027	A C D I K L P S T V W X Y Z
6	P	Rs 7,000	Rs 34,000	18 Jan 2027	A C D I K L M N O P Q S T V W X Y Z
7	I	Rs 10,500	Rs 44,500	15 Jan 2027	A C D I K L M N O P Q S T V W X Y Z
8	I	Rs 10,500	Rs 55,000	14 Jan 2027	A C D H I J K L M N O P Q S T V W X Y Z

Table 9: Crashing steps (each step saves 1 working day)

The first 5 days are cheap (Rs 27,000, from requirements and design). From step 6 the admin chain M - N - O - Q, which had only 1 day of float, also becomes critical, and from step 8 the chain H - J becomes critical too, so every further day needs more activities to be shortened at the same time. After 8 days no single or paired crash can shorten the project any more. The total crash cost of Rs 55,000 would come from the contingency reserve, which needs sponsor approval above Rs 50,000 per event.

8.2 Fast Tracking

Fast tracking overlaps activities that were planned in sequence, without extra cost but with a risk of rework. The best candidate is the discretionary part of K -> L: payment integration can start when the checkout screens are ready, before the coupon and delivery-charge rules are finished. Changing K FS -> L to K SS+9 -> L lets L start 5 days earlier, but go-live only moves to **25 Jan 2027** (1 working day earlier). The reason is the near-critical admin chain: tracking (P) also waits for courier integration (N), and testing (S) also waits for the reports dashboard (Q), so after 1 day the chain M - N - O - Q becomes critical. Fast tracking K and L alone is

therefore not worth the rework risk and the extra load on Developer 1; it only helps if the admin chain is also shortened.

8.3 Recommendation

The baseline keeps the original durations because the go-live date already meets the charter. If compression is required, crash C and D first (5 days for Rs 27,000) because these are planning-phase activities where extra people do not add technical risk. Fast tracking K and L is not recommended on its own, and crashing beyond 5 days is not recommended because the near-critical paths make it expensive and leave no float anywhere in the development phase.

9. BUDGET ESTIMATION

9.1 Estimating Method

A **bottom-up** estimate was used because the WBS dictionary already gives effort per role for every work package. For each activity:

Activity cost = Σ (person-days of role x daily rate of role) + non-labour cost

The cost elements are direct labour (internal team and contractors at daily rates that include their overheads) and direct non-labour costs (design tool licence, cloud hosting during the project, backup storage, external security audit, product photo shoot and printed manuals). Costs after go-live, such as gateway fees, SMS charges and hosting from April, are operating costs paid by ShopEase and are not part of the project budget. Example, activity I: 14 developer-days x Rs 3,500 = Rs 49,000.

9.2 Activity Cost Estimates

ID	Activity	WBS	Effort (person-days)	Labour	Non-labour	Total
A	Project initiation and charter	1.1.1	PM 5, BA 1	Rs 20,500	-	**Rs 20,500**
B	Project planning (baselines)	1.1.2	PM 6, BA 2	Rs 27,000	-	**Rs 27,000**
C	Requirements specification	1.2.1	BA 10, PM 2	Rs 37,000	-	**Rs 37,000**
D	UI/UX design and prototype test	1.2.2	UX 15, BA 2	Rs 51,000	Rs 6,000	**Rs 57,000**
E	Architecture and database design	1.2.3	DEV 8, OPS 2	Rs 35,600	-	**Rs 35,600**
F	Cloud hosting and environments	1.6.1	OPS 6	Rs 22,800	Rs 36,000	**Rs 58,800**
G	CI/CD pipeline	1.6.2	OPS 5	Rs 19,000	-	**Rs 19,000**
H	User accounts and OTP login	1.3.2	DEV 8	Rs 28,000	-	**Rs 28,000**
I	Product catalogue and search	1.3.1	DEV 14	Rs 49,000	-	**Rs 49,000**

ID	Activity	WBS	Effort (person-days)	Labour	Non-labour	Total
J	Product and inventory admin	1.5.1	DEV 10	Rs 35,000	-	**Rs 35,000**
K	Shopping cart and checkout	1.3.3	DEV 14	Rs 49,000	-	**Rs 49,000**
L	Payment gateway integration	1.4.1	DEV 10	Rs 35,000	-	**Rs 35,000**
M	Order management and returns	1.5.2	DEV 10	Rs 35,000	-	**Rs 35,000**
N	Shipping and courier integration	1.4.2	DEV 6	Rs 21,000	-	**Rs 21,000**
O	GST invoice generation	1.4.3	DEV 5, BA 1	Rs 20,500	-	**Rs 20,500**
P	Order tracking and notifications	1.3.4	DEV 8	Rs 28,000	-	**Rs 28,000**
Q	Sales reports dashboard	1.5.3	DEV 6	Rs 21,000	-	**Rs 21,000**
R	Security hardening and backups	1.6.3	OPS 6	Rs 22,800	Rs 8,000	**Rs 30,800**
S	Functional and integration testing	1.7.1	QA 18, DEV 4	Rs 64,400	-	**Rs 64,400**
T	Performance and security testing (VAPT)	1.7.2	QA 6, OPS 3	Rs 28,200	Rs 35,000	**Rs 63,200**
U	Product data migration	1.8.1	BA 6, DEV 3	Rs 28,500	Rs 30,000	**Rs 58,500**
V	User acceptance testing	1.7.3	QA 4, BA 4, PM 2	Rs 30,200	-	**Rs 30,200**
W	Staff training and user manual	1.8.2	BA 5	Rs 15,000	Rs 5,000	**Rs 20,000**
X	Go-live preparation and deployment	1.8.3	DEV 2, OPS 2, QA 1, PM 1	Rs 20,900	-	**Rs 20,900**
Y	Hypercare (2 weeks)	1.8.3	DEV 4, OPS 2, QA 1, PM 2	Rs 31,400	-	**Rs 31,400**
Z	Project closure	1.1.4	PM 4, BA 1	Rs 17,000	-	**Rs 17,000**
LOE	Monitoring and status reporting	1.1.3	PM 30	Rs 1,05,000	-	**Rs 1,05,000**
	Total (work packages)		**267 person-days**	**Rs 8,97,800**	**Rs 1,20,000**	**Rs 10,17,800**

Table 10: Bottom-up cost estimate per activity

9.3 Cost by Resource and Element

Role / element	Rate (Rs/day)	Person-days	Cost
Project Manager	3500	52	Rs 1,82,000
Business Analyst	3000	32	Rs 96,000
UI/UX Designer	3000	15	Rs 45,000
Developers	3500	112	Rs 3,92,000
QA Engineer	2800	30	Rs 84,000
DevOps Engineer	3800	26	Rs 98,800

Role / element	Rate (Rs/day)	Person-days	Cost
Non-labour (licence, hosting, backup storage, VAPT, photo shoot, manuals)			Rs 1,20,000
Total			**Rs 10,17,800**

Table 11: Cost by role

9.4 Reserves

Two different reserves are added on top of the work package costs:

- **Contingency reserve** is for **identified risks** ("known unknowns") from the risk register. It is part of the cost baseline, is controlled by the project manager within the limits in the charter and is used when a listed risk happens.
- **Management reserve** is for **unidentified risks** ("unknown unknowns"). It is not part of the cost baseline, is held by the sponsor and needs a change to the baseline when it is used.

The contingency reserve was checked against the expected monetary value (EMV = probability x cost impact) of the risks in the charter:

ID	Risk	Probability	Cost impact	EMV
R1	Payment gateway KYC delayed	30%	Rs 60,000	Rs 18,000
R2	Product data and photos incomplete or late	50%	Rs 40,000	Rs 20,000
R3	Store staff resist the new process (extra training)	30%	Rs 30,000	Rs 9,000
R4	Approved scope changes during development	40%	Rs 80,000	Rs 32,000
R5	A developer leaves (replacement and handover)	10%	Rs 1,20,000	Rs 12,000
R6	Site slows down at launch (extra servers)	10%	Rs 50,000	Rs 5,000
R7	Security incident (audit and fixes)	5%	Rs 1,20,000	Rs 6,000
	Total EMV			**Rs 1,02,000**

Table 12: Expected monetary value of the identified risks

The total EMV is Rs 1,02,000, which is the same as 10% of the work package cost (Rs 1,02,000), so 10% is used as the contingency reserve. The management reserve is set at 5% (Rs 51,000), as agreed with the sponsor in the charter.

9.5 Budget Aggregation

Level	Item	Amount
Control account	1.1 Project Management	Rs 1,69,500
Control account	1.2 Requirements and Design	Rs 1,29,600
Control account	1.3 Customer Storefront	Rs 1,54,000
Control account	1.4 Payments and Integrations	Rs 76,500
Control account	1.5 Admin and Back Office	Rs 91,000

Level	Item	Amount
Control account	1.6 Infrastructure and Deployment	Rs 1,08,600
Control account	1.7 Testing and Quality	Rs 1,57,800
Control account	1.8 Launch and Transition	Rs 1,30,800
	Work package total = performance measurement baseline (PMB)	**Rs 10,17,800**
+	Contingency reserve (10%)	Rs 1,02,000
=	**Cost baseline**	**Rs 11,19,800**
+	Management reserve (5%)	Rs 51,000
=	**Total project budget**	**Rs 11,70,800**

Table 13: Cost aggregation from control accounts to total budget

The totals are the same as in the charter of Assignment 1, which confirms that the budget is built from the WBS work packages and not set top-down.

10. TIME-PHASED COST BASELINE AND S-CURVE

10.1 Method

The cost of each activity was spread evenly over its working days and then added up per calendar week (week 1 starts on Monday 31 Aug 2026; work starts on Tuesday 01 Sep 2026). This was done twice: once with the early start dates and once with the late start dates from the CPM. The monitoring activity (level of effort) is spread evenly from week 2 to week 24. The contingency reserve is distributed in proportion to the planned work, so the cost baseline curve is the planned value curve x 1.10.

Week	Starts	Weekly PV (early)	Cumulative PV (early)	Weekly PV (late)	Cumulative PV (late)	Cumulative cost baseline	% complete
1	31 Aug	Rs 16,400	Rs 16,400	Rs 16,400	Rs 16,400	Rs 18,044	2%
2	07 Sep	Rs 41,186	Rs 57,586	Rs 23,186	Rs 39,586	Rs 63,357	6%
3	14 Sep	Rs 28,086	Rs 85,671	Rs 37,086	Rs 76,671	Rs 94,257	8%
4	21 Sep	Rs 41,957	Rs 1,27,629	Rs 33,157	Rs 1,09,829	Rs 1,40,419	13%
5	28 Sep	Rs 50,526	Rs 1,78,154	Rs 26,606	Rs 1,36,434	Rs 1,96,008	18%
6	05 Oct	Rs 92,117	Rs 2,70,271	Rs 42,157	Rs 1,78,591	Rs 2,97,357	27%
7	12 Oct	Rs 45,057	Rs 3,15,329	Rs 37,937	Rs 2,16,529	Rs 3,46,930	31%
8	19 Oct	Rs 32,286	Rs 3,47,614	Rs 32,286	Rs 2,48,814	Rs 3,82,451	34%
9	26 Oct	Rs 40,357	Rs 3,87,971	Rs 40,357	Rs 2,89,171	Rs 4,26,852	38%
10	02 Nov	Rs 39,097	Rs 4,27,069	Rs 89,957	Rs 3,79,129	Rs 4,69,868	42%
11	09 Nov	Rs 15,303	Rs 4,42,371	Rs 40,543	Rs 4,19,671	Rs 4,86,704	43%
12	16 Nov	Rs 55,757	Rs 4,98,129	Rs 44,157	Rs 4,63,829	Rs 5,48,049	49%
13	23 Nov	Rs 40,357	Rs 5,38,486	Rs 40,357	Rs 5,04,186	Rs 5,92,451	53%
14	30 Nov	Rs 74,392	Rs 6,12,878	Rs 50,017	Rs 5,54,203	Rs 6,74,298	60%

Week	Starts	Weekly PV (early)	Cumulative PV (early)	Weekly PV (late)	Cumulative PV (late)	Cumulative cost baseline	% complete
15	07 Dec	Rs 83,232	Rs 6,96,110	Rs 70,577	Rs 6,24,780	Rs 7,65,871	68%
16	14 Dec	Rs 66,807	Rs 7,62,917	Rs 87,682	Rs 7,12,462	Rs 8,39,374	75%
17	21 Dec	Rs 58,366	Rs 8,21,283	Rs 84,446	Rs 7,96,908	Rs 9,03,589	81%
18	28 Dec	Rs 0	Rs 8,21,283	Rs 0	Rs 7,96,908	Rs 9,03,589	81%
19	04 Jan	Rs 54,517	Rs 8,75,800	Rs 78,892	Rs 8,75,800	Rs 9,63,569	86%
20	11 Jan	Rs 35,524	Rs 9,11,324	Rs 35,524	Rs 9,11,324	Rs 10,02,653	90%
21	18 Jan	Rs 38,565	Rs 9,49,889	Rs 38,565	Rs 9,49,889	Rs 10,45,084	93%
22	25 Jan	Rs 21,431	Rs 9,71,320	Rs 21,431	Rs 9,71,320	Rs 10,68,662	95%
23	01 Feb	Rs 21,057	Rs 9,92,377	Rs 21,057	Rs 9,92,377	Rs 10,91,829	98%
24	08 Feb	Rs 18,623	Rs 10,11,000	Rs 18,623	Rs 10,11,000	Rs 11,12,319	99%
25	15 Feb	Rs 6,800	Rs 10,17,800	Rs 6,800	Rs 10,17,800	Rs 11,19,800	100%

Table 14: Time-phased planned value and cost baseline by week

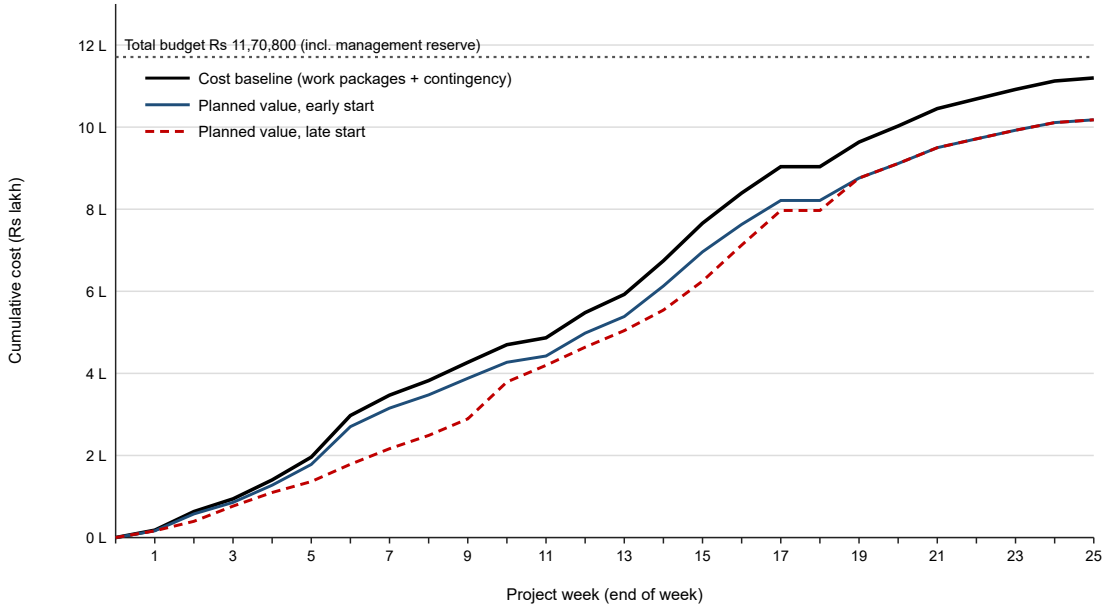


Figure 4: S-curve of the cost baseline with early and late start planned value and the total budget

10.2 Reading the S-Curve

- The curve is flat in the first weeks (planning and requirements need few people), steep in weeks 7 to 19 (two developers, QA and DevOps working together) and flat again during hypercare and closure. It also flattens in the holiday weeks: Diwali in week 11 and the year-end closure in week 18.
- The area between the early start and late start curves is the cost effect of float. Spending that stays between the two curves is still on schedule; spending below the late start curve means the project is behind schedule.

- The cost baseline ends at Rs 11,19,800. The gap between the end of the cost baseline and the total budget line (Rs 11,70,800) is the management reserve, which is not part of the baseline.
- The cumulative planned value is the PV used later in earned value management, where it will be compared with earned value (EV) and actual cost (AC) to calculate SPI and CPI.

10.3 Funding Requirement by Month

Month	Planned spend (cost baseline)	Cumulative
Sep 2026	Rs 1,96,008	Rs 1,96,008
Oct 2026	Rs 2,30,844	Rs 4,26,852
Nov 2026	Rs 1,65,598	Rs 5,92,451
Dec 2026	Rs 3,11,138	Rs 9,03,589
Jan 2027	Rs 1,65,073	Rs 10,68,662
Feb 2027	Rs 51,138	Rs 11,19,800

Table 15: Monthly funding requirement

The owner releases funds monthly in line with this table. The management reserve is kept aside by the owner and released only through change control.

11. DYNAMIC SCHEDULE AND BUDGET MODEL

All calculations in this document are also built as formulas in the workbook **ShopEase_Schedule_Budget_Model.xlsx**, so the model is not a static picture. The workbook has these sheets:

Sheet	Contents
Inputs	Project start date, holidays, daily rates, contingency % and management reserve % (yellow input cells)
Links	All 37 dependencies with type (FS, SS, FF) and lag; type and lag can be edited
CPM	O, M, P estimates, PERT duration, forward pass, backward pass, total and free float, critical flag, dates, project finish, go-live date, critical path text and probability
Costs	Effort per role and non-labour cost per activity, labour cost, total cost and cost per day
Budget	Roll-up to control accounts, PMB, contingency, cost baseline, management reserve and total budget, with a check against the Costs sheet
S-Curve	Weekly time-phased cost for early and late start, cumulative curves and the S-curve chart
Compression	Crash durations, extra costs and cost slopes

Table 16: Workbook structure

Each ES cell is a MAX formula over its predecessors that reads the dependency type and lag from the Links sheet, for example for activity H:

=MAX(0, IF(type="FS", EF_E+lag, IF(type="SS", ES_E+lag, EF_E+lag-D_H)), IF(type="SS", ES_D+lag, ...))

The LF cells use MIN formulas over successors in the same way, dates use WORKDAY with the holiday list, and the weekly cost uses NETWORKDAYS between the activity dates and the week dates. Changing any input recalculates the floats, critical path, dates, budget and S-curve. The workbook was opened in Microsoft Excel and its results match every table in this report. The following what-if tests show how the model reacts:

What-if change	Duration (days)	Go-live	Critical path	Explanation
Baseline	108	27 Jan 2027	A C D I K L P S T V W X Y Z	-
Payment gateway (L) takes 14 days instead of 10 (KYC delay, risk R1)	112	02 Feb 2027	Same	L is critical, so go-live moves by 4 days
Data migration (U) takes 17 days instead of 12	108	27 Jan 2027	Same	U has 12 days of float, so nothing moves
Reports dashboard (Q) takes 8 days instead of 6	109	28 Jan 2027	A C D I K M N O Q S T V W X Y Z	Q had only 1 day of float; the extra day delays go-live and the admin chain becomes critical
Security work (R) takes 32 days instead of 10	109	28 Jan 2027	A C E F G R T V W X Y Z	R had 21 days of float; a slip of 22 days uses it all and delays go-live by 1 day through T

Table 17: What-if tests on the dynamic model

When an activity moves, its cost moves to other weeks, so the S-curve changes shape while the total budget stays the same. A change in a rate or in the contingency percentage changes the budget totals and the height of the curve.

12. CONCLUSION

The 25 work packages from Assignment 1 were turned into 26 activities with PERT durations and 37 dependencies, using FS, SS and FF relationships with leads and lags. The CPM gives a project duration of 108 working days with the critical path A - C - D - I - K - L - P - S - T - V - W - X - Y - Z, go-live on 27 Jan 2027 and closure on 16 Feb 2027, which confirms the charter. The calculation also showed a near-critical admin chain with only 1 day of float, resource over-allocations that were levelled

within float, and that the cheapest way to gain one week is to crash requirements and design for Rs 27,000.

The bottom-up budget rolls up from activities to control accounts and gives a work package total of Rs 10,17,800. A contingency reserve supported by an EMV analysis gives a cost baseline of Rs 11,19,800, and a management reserve gives a total budget of Rs 11,70,800. The cost baseline was time-phased and plotted as an S-curve for early and late start schedules. The Excel model makes the schedule and the budget dynamic, so the plan can be updated quickly when estimates or dependencies change during the project.

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